

1.1.1 – 1.1.2 Traditional vs Natural Interaction

روایتی تعامل (Traditional Interaction) میں ماؤس، کی بورڈ، کی پیڈ یا عام ٹیپ اور سوائپ استعمال ہوتے ہیں، جیسے اے ٹی ایم کی پیڈ۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 02 of 15

1.1.1 – 1.1.2 Traditional vs Natural Interaction

LEARN IT

1.1.1 Traditional interaction is the basic way people use everyday devices. It relies on standard input and output: a mouse, a keyboard, a keypad, a monitor, speakers, or a simple touchscreen tap and swipe.

- **Computer:** moving a mouse and typing on a keyboard; output via monitor and speakers.
- **Smartphone (traditional use):** tapping and swiping on a touchscreen; output via screen lighting and vibrations.

1.1.2 Natural interaction is advanced, smart technology that lets people interact **without a mouse or keyboard**. It feels more human — you speak, move, or show your face, and the system understands.

- **Voice:** speaking commands into a device (for example, "Alexa, play a song").
- **Body and gesture:** facial recognition to unlock a phone, or playing VR games by waving hands.

Feature	Traditional	Natural
Main input	Mouse, keyboard, keypad, simple tap/swipe	Voice, face, hand/body gesture, movement
Feels like	Operating a machine	Talking or moving as you do with people
Examples	TV remote, ATM keypad, school-portal login typed on a PC	FaceID, Siri, Kinect games, "Hey Google"
Typical output	Screen, speakers, vibration	Spoken reply, unlock, tracking of the body in VR

WORKED EXAMPLE

Worked example: sort today's tools. List three traditional and three natural interfaces you might use in one day in Sindh.

Traditional: (1) HBL ATM keypad to withdraw cash. (2) College computer lab: mouse + keyboard to open the school portal. (3) Television remote at home.

Natural: (1) Unlocking a phone with FaceID. (2) Asking Google Assistant "Call Aslam". (3) A VR science demo in which you pick a molecule by waving your hand.

Notice: the *same smartphone* can be traditional (you type a PIN) or natural (the camera reads your face). The classification depends on **how you interact**, not on the brand of the device.

REMEMBER IT

Test sentence: If you must press keys or tap labelled buttons, it is traditional. If the machine understands speech, face or body movement, it is natural.

COMMON MISTAKE

Mistake: "Touchscreens are always natural interaction."

Fix: Ordinary tapping and swiping on a phone is still listed as traditional in this chapter. Natural interaction starts when the system understands voice, face, or full-body gesture without a mouse or keyboard.

اردو میں سمجھیں

روایتی تعامل (Traditional Interaction) میں ماؤس، کی بورڈ، کی پیڈ یا عام ٹیپ اور سوائپ استعمال ہوتے ہیں، جیسے اے ٹی ایم کی پیڈ۔ قدرتی تعامل (Natural Interaction) میں آواز، چہرہ یا جسم کی حرکت سے کام ہوتا ہے، جیسے فیس آئی ڈی (FaceID) یا "الیکسا، گانا چلاؤ"۔ درجہ بندی آلے کی نہیں، تعامل کے طریقے کی ہوتی ہے۔

CHECK YOURSELF

1. Is an ATM keypad traditional or natural?
2. Give one voice example of natural interaction from the source.
3. Why can one smartphone be both traditional and natural?

Answers: 1. Traditional. 2. "Alexa, play a song" (or any spoken command). 3. Because typing a PIN is traditional, while FaceID or a voice assistant on the same phone is natural — it depends on the method, not the device.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Traditional interaction uses:

- a. Standard controls — keys, mouse, menus
- b. Gaze and gesture as the default 1980 PC
- c. Smell
- d. Telepathy

MCQ 2. Natural interaction tries to:

- a. Use human skills (speech, gesture, touch) with less training
- b. Force people to learn command codes only
- c. Remove all feedback
- d. Use only punch cards

MCQ 3. A command-line is more:

- a. Traditional / learned
- b. Natural baby-talk
- c. IoT soil
- d. A pie

MCQ 4. Pinch-to-zoom on a phone is more:

- a. Natural (gesture)
- b. A foreign key
- c. A K-map
- d. Waterfall

MCQ 5. Both still need:

- a. Feedback so the user knows the system heard them
- b. No UI
- c. No errors ever
- d. No accessibility

MCQ 6. Alexa-style speech is:

- a. Natural-ish voice interaction
- b. A datasheet
- c. A histogram
- d. A stack pop

Extra short questions

1. Define traditional vs natural interaction.
2. One example each in a college.
3. A drawback of “natural”.
4. Can a product mix both?

Extra long question

1. Design a library search: one traditional UI and one natural UI. Compare learnability, errors, and accessibility.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Traditional: trained widgets (keyboard/mouse). Natural: speech, gesture, touch that feel like the real world.
2. Traditional: SIS forms. Natural: fingerprint attendance, touch kiosk.
3. False gesture recognition; privacy of voice; not natural for every disability.
4. Yes — a phone has icons (traditional GUI) and gestures/voice.

Long

1. Traditional: typed query + filters. Natural: speak the title / point at a shelf QR.
Learnability vs noisy halls. Accessibility: keyboard vs voice. Recommend both.

Next lecture: 1.2 Applications of HCI (Domains of Life)